

CM0889 Analysis of Algorithms

Appendix

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Induction

Principle of Mathematical Induction

Let $P(n)$ be a property on natural numbers n , and let a be a fixed natural number.

If

(i) $P(a)$ is true and

(ii) for every natural number $k \geq a$, if $P(k)$ is true then $P(k + 1)$ is true,

then

(iii) $P(n)$ is true for all natural numbers $n \geq a$.

Induction

Principle of Strong Induction

Let $P(n)$ be a property on natural numbers n , and let a be a fixed natural number.

If

- (i) $P(a)$ is true and
 - (ii) for every natural number $k \geq a$, if $P(i)$ is true for $a \leq i \leq k$, then $P(k+1)$ is true,
- then
- (iii) $P(n)$ is true for all natural numbers $n \geq a$.

Remark

Other names: Principle of complete induction and principle of course-of-values induction.

Induction

Theorem

The principle of mathematical induction and the principle of strong induction are equivalents.

Floor and Ceiling Functions

Definition

The **floor** function is defined by

$$\lfloor \cdot \rfloor : \mathbb{R} \rightarrow \mathbb{Z}$$

$$\lfloor x \rfloor := \text{largest integer less than or equal to } x.$$

Example

$$\lfloor 42 \rfloor = 42,$$

$$\lfloor 5.42 \rfloor = 5,$$

$$\lfloor -5.52 \rfloor = -6.$$

Floor and Ceiling Functions

Definition

The **ceiling** function is defined by

$$\lceil \cdot \rceil : \mathbb{R} \rightarrow \mathbb{Z}$$

$$\lceil x \rceil := \text{smallest integer greater than or equal to } x.$$

Example

$$\lceil 42 \rceil = 42,$$

$$\lceil 5.42 \rceil = 6,$$

$$\lceil -5.52 \rceil = -5.$$

Logarithms

Definition

For any fixed real number $b > 1$,

$$\begin{aligned}\log_b : \mathbb{R}^+ &\rightarrow \mathbb{R} \\ \log_b x = y &\quad \text{iff} \quad b^y = x.\end{aligned}$$

Logarithms

Definition

For any fixed real number $b > 1$,

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Notation

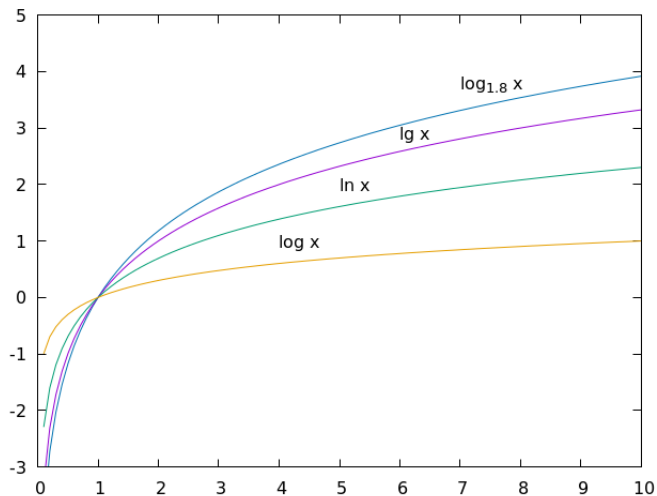
$\lg x$: Logarithm on base 2

$\ln x$: Logarithm on base e

$\log x$: Logarithm on base 10

Logarithms

Logarithmic functions $\lg x$, $\ln x$, $\log x$ and $\log_{1.8} x$



Logarithms

Properties

For any fixed real number $b > 1$, for all $x, y \in \mathbb{R}^+$, and for all $z \in \mathbb{R}$:

$$\log_b(xy) = \log_b x + \log_b y,$$

$$\log_b(x/y) = \log_b x - \log_b y,$$

$$\log_b(x^z) = z \log_b x.$$

Logarithms

Properties

For all real numbers a and b greater than 1 and for all $x \in \mathbb{R}^+$:

$$\log_b x = \frac{\log_a x}{\log_a b}.$$

Logarithms

Properties

For all real numbers a and b greater than 1, for all $x \in \mathbb{R}^+$, if $a < b$ then

$$\log_a x > \log_b x.$$

Summations

Definition

Let $a_1, a_2, \dots, a_n$ be a sequence of numbers, where n is a positive integer. Recall the recursive definition of the **summation notation**:

$$\begin{aligned}\sum_{k=1}^1 a_k &:= a_1, \\ \sum_{k=1}^n a_k &:= \left(\sum_{k=1}^{n-1} a_k \right) + a_n \\ &= a_1 + a_2 + \cdots + a_{n-1} + a_n.\end{aligned}$$

Summations

Properties

$$\sum_{k=1}^n (a_k + b_k) = \sum_{k=1}^n a_k + \sum_{k=1}^n b_k$$

(additive property),

$$\sum_{k=1}^n ca_k = c \sum_{k=1}^n a_k$$

(homogeneous property),

$$\sum_{k=1}^n (\alpha a_k + \beta b_k) = \alpha \sum_{k=1}^n a_k + \beta \sum_{k=1}^n b_k$$

(linearity property).

Summations

Properties

$$\sum_{k=1}^n f(n) = nf(n),$$

$$\sum_{k=1}^n a_k = \sum_{k=1}^i a_k + \sum_{k=i+1}^n a_k.$$

Summations

Properties

$$\sum_{k=1}^n k = \frac{n(n+1)}{2},$$

$$\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6},$$

$$\sum_{k=1}^n k^3 = \left(\frac{n(n+1)}{2} \right)^2.$$